

Unit 2: Introduction to CAD

Quiz #1 Study Guide

Fusion 360 Learning Check

Where are your Fusion 360 projects automatically saved by default?

This enables easy access across devices and automatic backups, as demonstrated in the project setup workflow.



Fusion 360 Learning Check

True or False Fusion 360's basic modeling features require a constant internet connection.

Basic modeling works offline, but cloud features like rendering or collaboration require internet.

Fusion 360 Learning Check

Which panel in the Fusion 360 interface shows the history of your design changes?

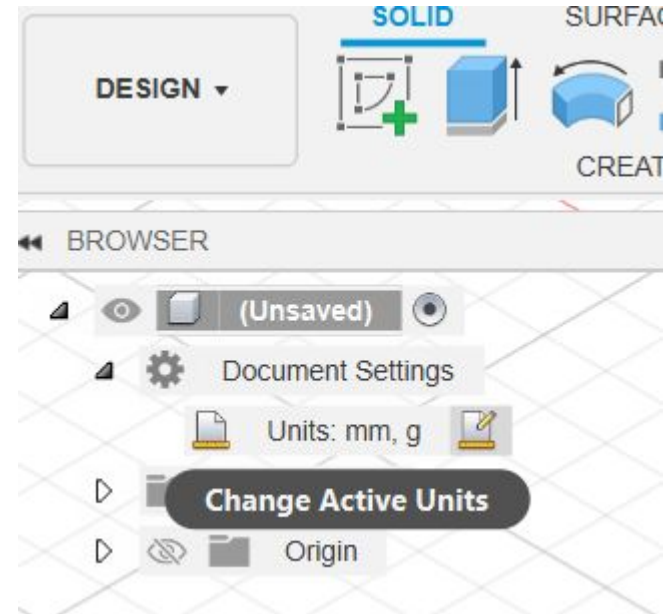
- a) Browser
- b) Toolbar
- c) Canvas
- d) Timeline



Fusion 360 Learning Check

How do you change units from inches to millimeters in Fusion 360?

Go to the Browser >
Document Settings >
Units, and select
"mm".



Fusion 360 Learning Check

What keyboard shortcut is used to orbit the view in the canvas?

a) Shift + Middle Mouse Button

b) Ctrl + Z

c) Alt + Drag

d) Spacebar

View controls like orbit, zoom (scroll wheel), and pan (middle mouse) are essential for navigation.

Fusion 360 Learning Check

What tool is used to draw a basic rectangle in a sketch?

a) Line

b) Circle

c) Rectangle

d) Spline

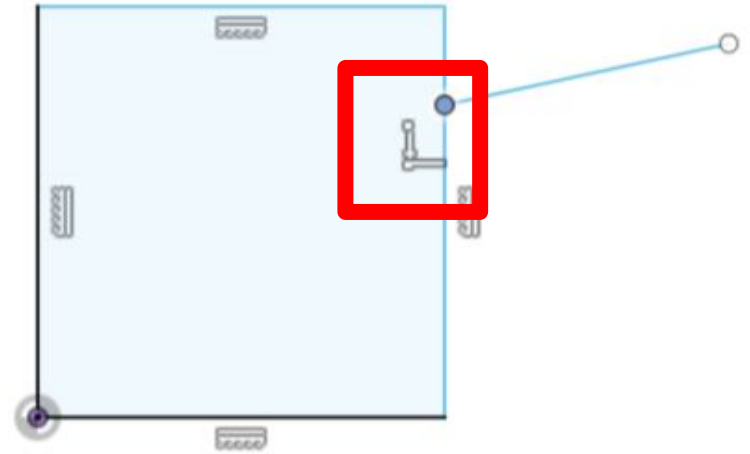
The Rectangle tool creates closed profiles quickly on a selected plane.

Fusion 360 Learning Check

What does a "coincident" constraint do in a sketch?

Answer: It forces two points to share the same location.

Explanation: Constraints like coincident ensure sketches are fully defined and editable.



Fusion 360 Learning Check

True or False: Sketches must be fully constrained (blue lines turn black) before extruding.

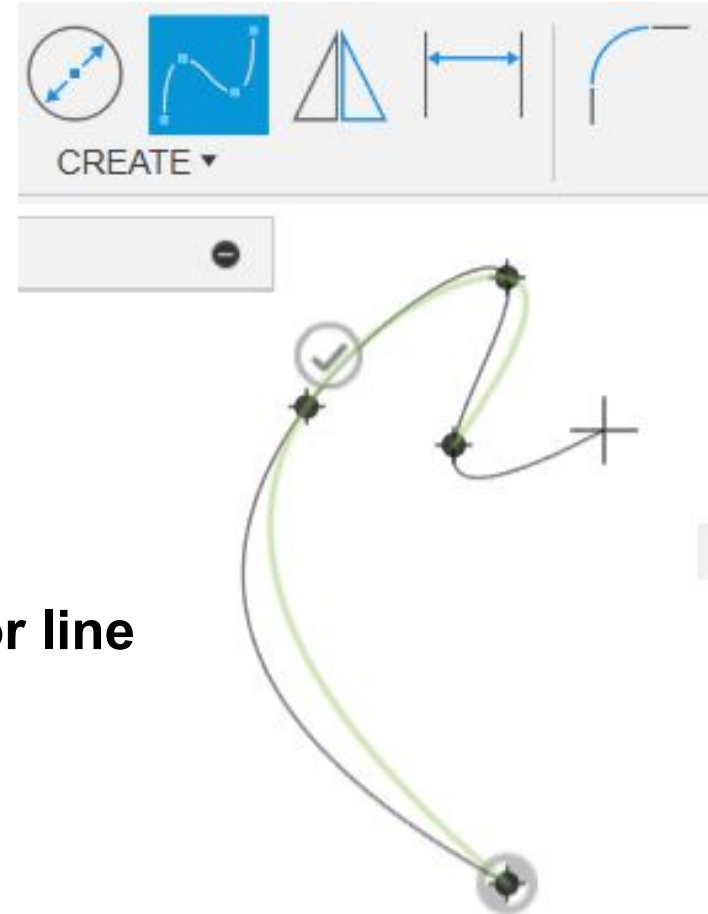
False. This is not true! It is recommended but not required.

Explanation: Under-constrained sketches (blue) can lead to unpredictable changes, as warned in the video.

Fusion 360 Learning Check

Which tool allows drawing freeform curves in a sketch?

- a) Arc **Creates a fixed radius arc**
- b) Polygon **Creates a multi-side polygon**
- c) Spline** **Splines are for organic shapes**
- d) Mirror **Copies geometry across a mirror line**

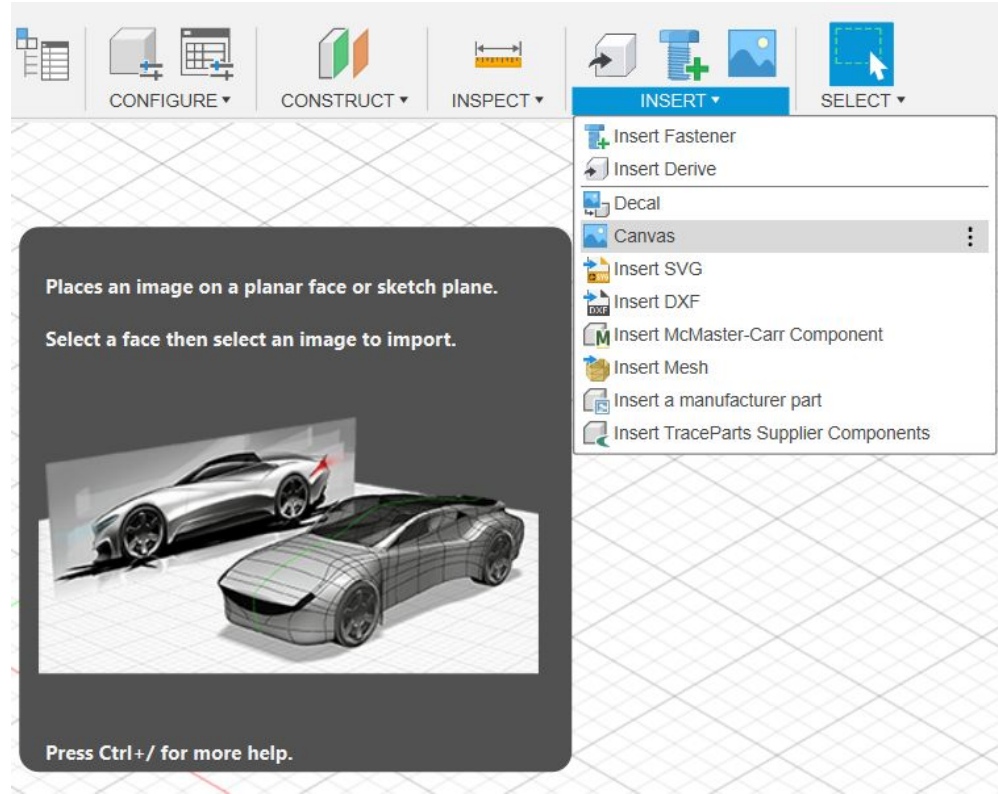


Fusion 360 Learning Check

How do you import an image into a sketch for tracing?

Answer: Insert > Canvas > Attach an image file.

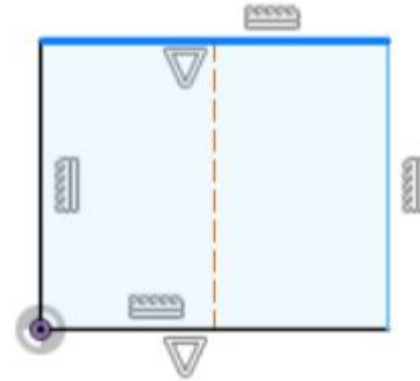
Canvases act as reference for dimensions, useful for reverse engineering.



Fusion 360 Learning Check

What is the purpose of construction lines in a sketch?

- a) To create solid geometry
- b) For reference only (not extruded)**
- c) To add color
- d) To delete features

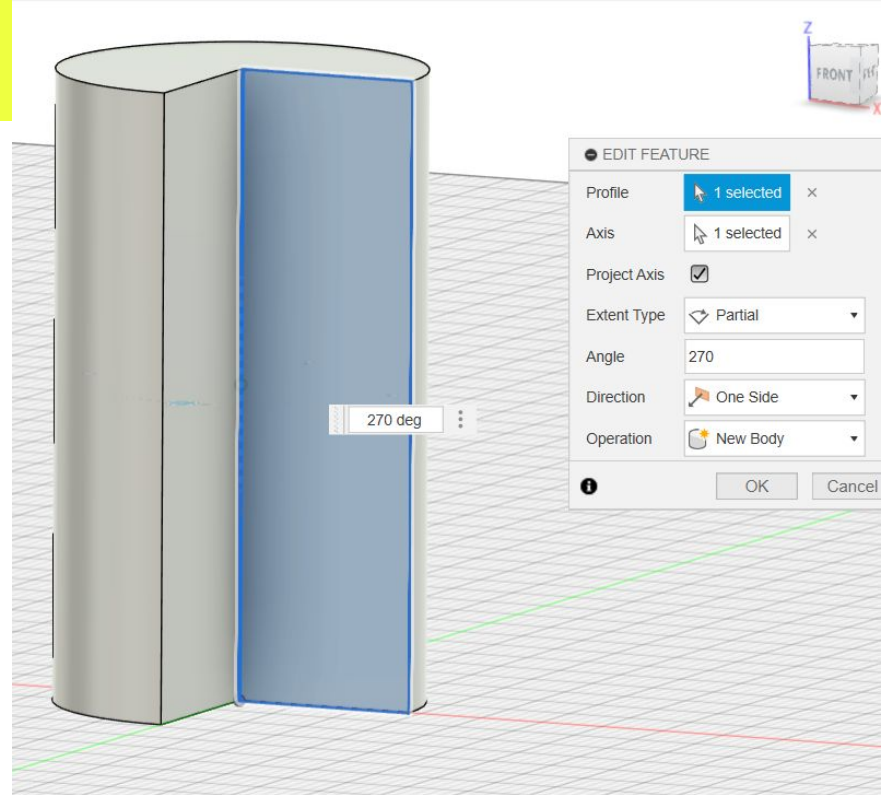


They help align elements without affecting the final solid elements.

Fusion 360 Learning Check

What operation can turn a 2D rectangle into a 3D solid cylinder-like shape?

- a) Fillet
- b) Revolve**
- c) Extrude
- d) Sweep



Explanation: Revolve spins a profile 360 degrees around an axis.

Fusion 360 Learning Check

In the timeline, how do you edit an extrude distance after creation?

Double-click the Extrude feature in the timeline and adjust the parameter.

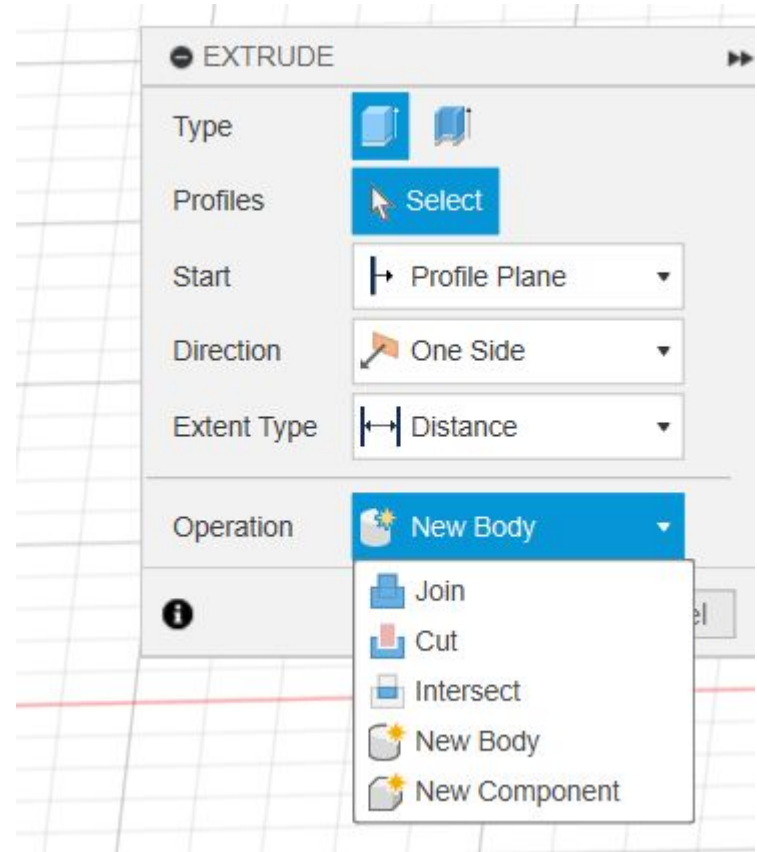
Parametric editing keeps designs flexible.



Fusion 360 Learning Check

When extruding, which boolean combines two bodies into one?

Join. Explanation: Other options like Cut or Intersect are for subtraction or overlap.



Fusion 360 Learning Check

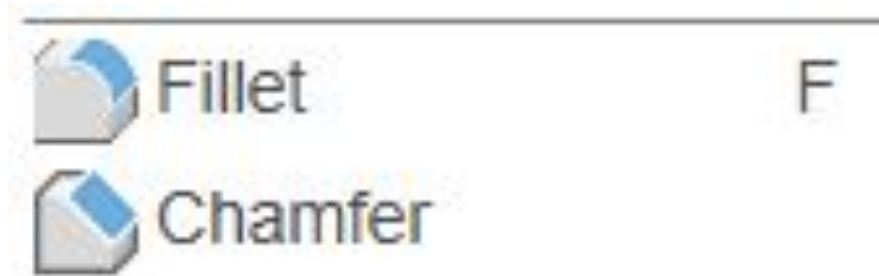
What tool rounds sharp edges on a model?

a) Chamfer

b) Hole

c) Fillet

d) Shell



Explanation: Fillets add curves; chamfers add bevels.

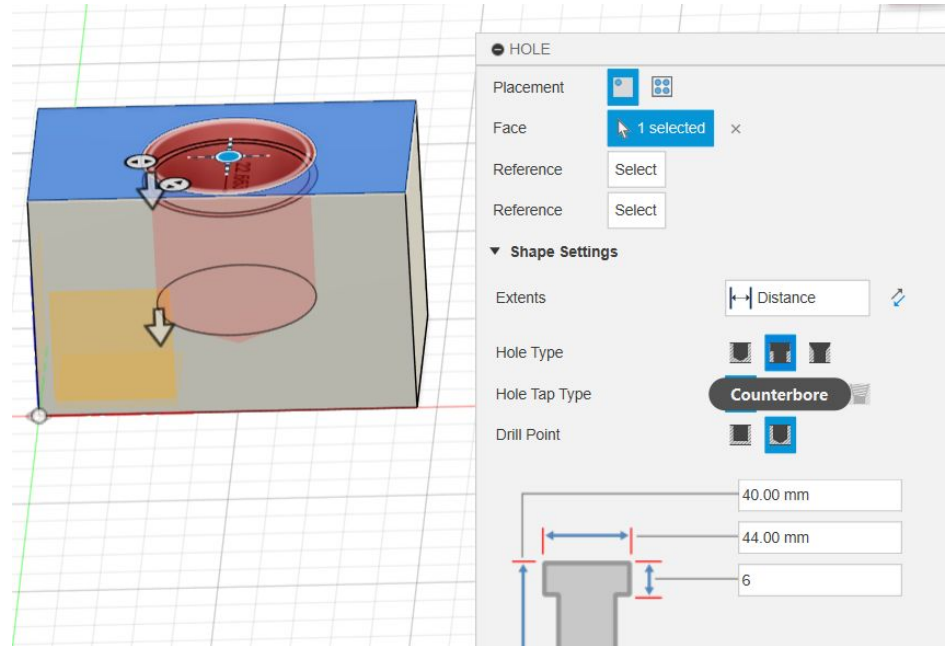
Fusion 360 Learning Check

Which hole type includes a recess for a screw head?

Counterbored.

Explanation:

Countersunk is for tapered screws; simple is straight through.



Fusion 360 Learning Check

True or False: Fillets can be applied to edges in a sketch before extruding.

Explanation: Fillets may be applied in a sketch or post-extrusion, but it is considered a best practice to apply them to 3d bodies.



Fillet

Places an arc of a specified radius at the intersection of two lines or arcs.

Select the vertex or the two lines or arcs. Specify a radius for the fillet.

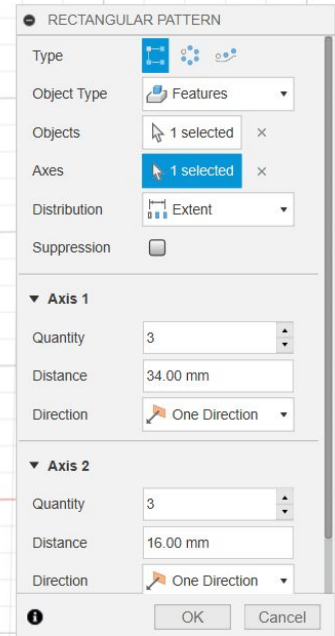
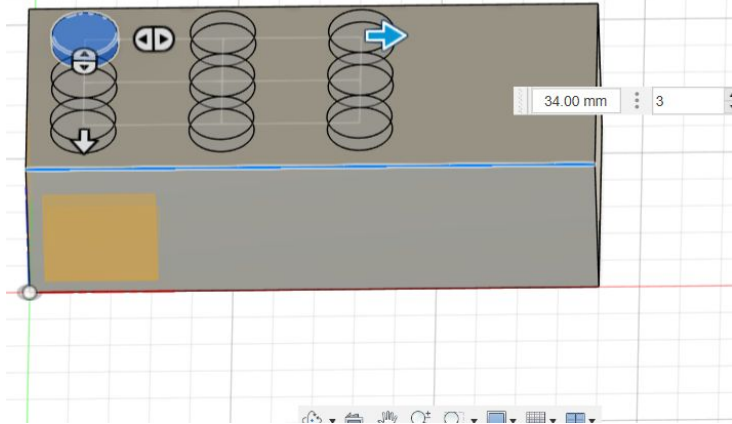
Press Ctrl+/ for more help.



Fusion 360 Learning Check

What creates multiple copies of a feature in a straight line?

- a) Circular Pattern
- b) Mirror
- c) Rectangular Pattern**
- d) Loft



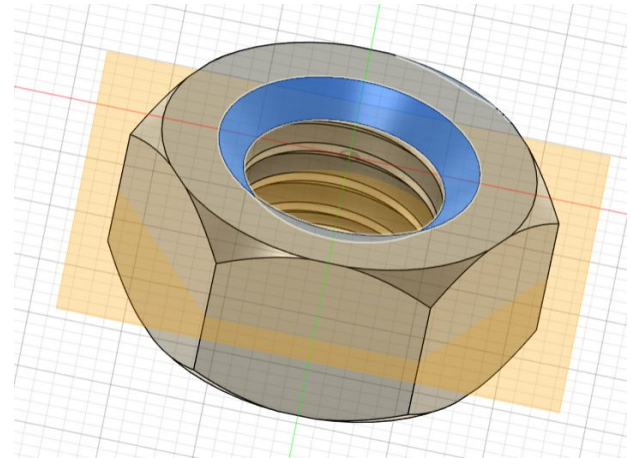
Explanation: Circular pattern copies geometry in a circle around an axis. Mirror creates one copy across a mirror line. Loft creates a solid from two or more sketches on different planes.

Fusion 360 Learning Check

How do you mirror a body across a plane?

Create > Mirror > Select body and mirror plane.

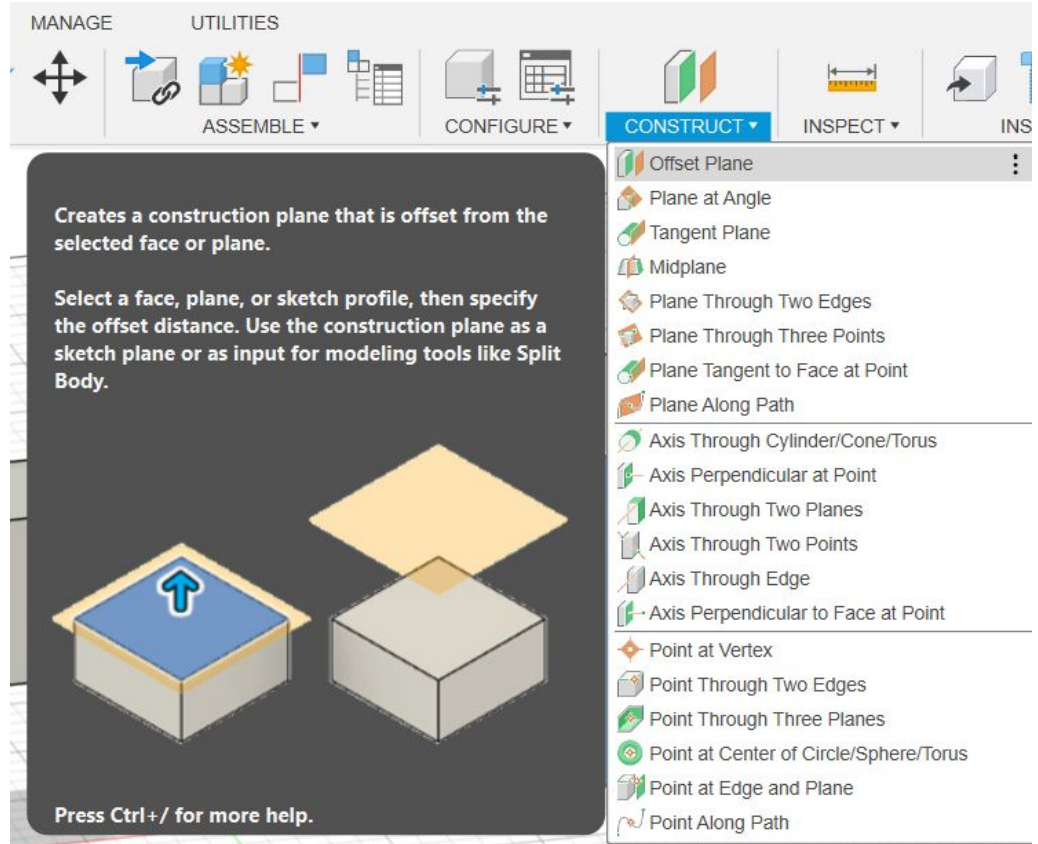
Useful for symmetrical designs like brackets.



Fusion 360 Learning Check

What is an offset plane used for?

To create a parallel plane for patterning on multi-body parts. Explanation: Helps position features away from origins.

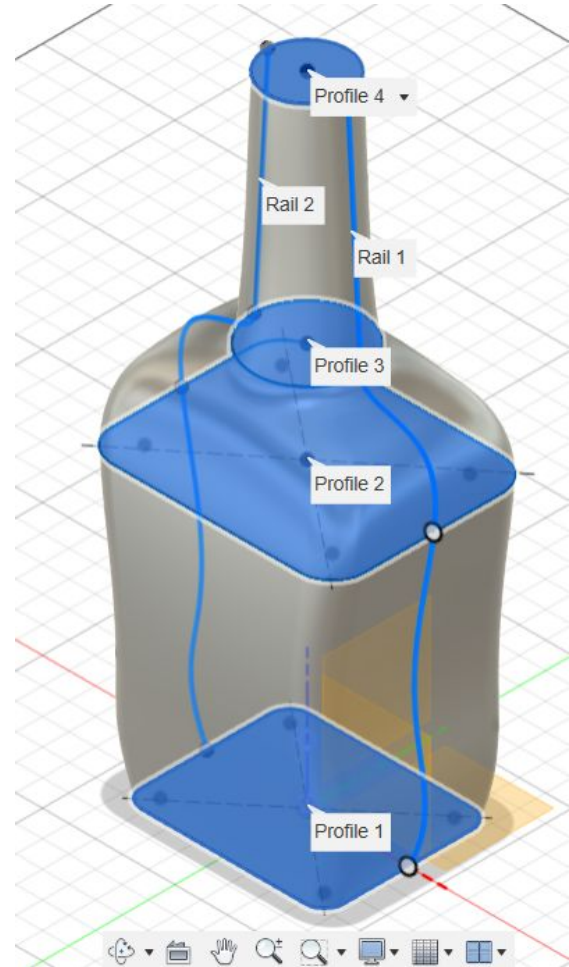


Fusion 360 Learning Check

Which feature blends between two profiles?

- a) Sweep
- b) Extrude
- c) Loft**
- d) Revolve

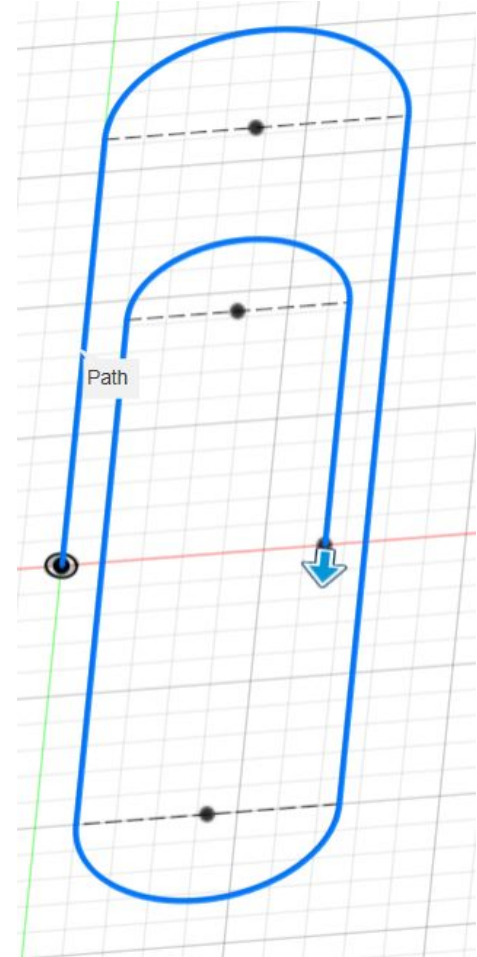
Lofts use rails for smooth transitions.



Fusion 360 Learning Check

What do you use to guide a profile along a path when using the sweep tool?

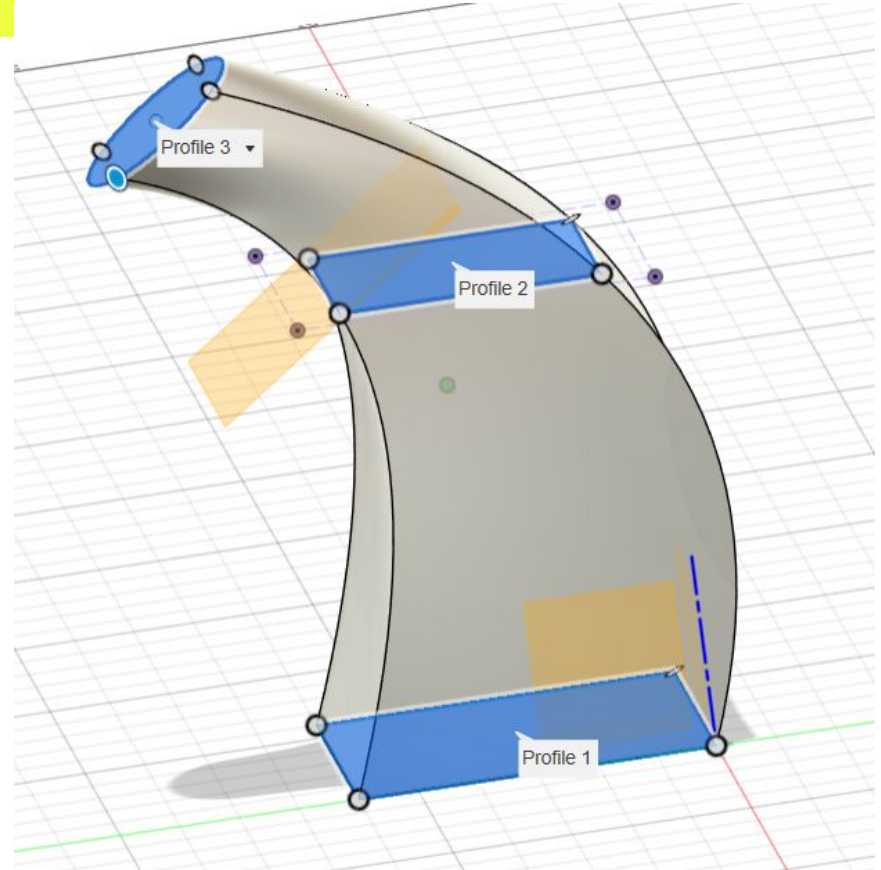
A continuous line or curve, i.e. from another sketch. In the phone case exercise, the path was derived from a continuous line in the phone body.



Fusion 360 Learning Check

True or **False:** Lofts require profiles on parallel planes only.

They work on any planes.



Fusion 360 Learning Check

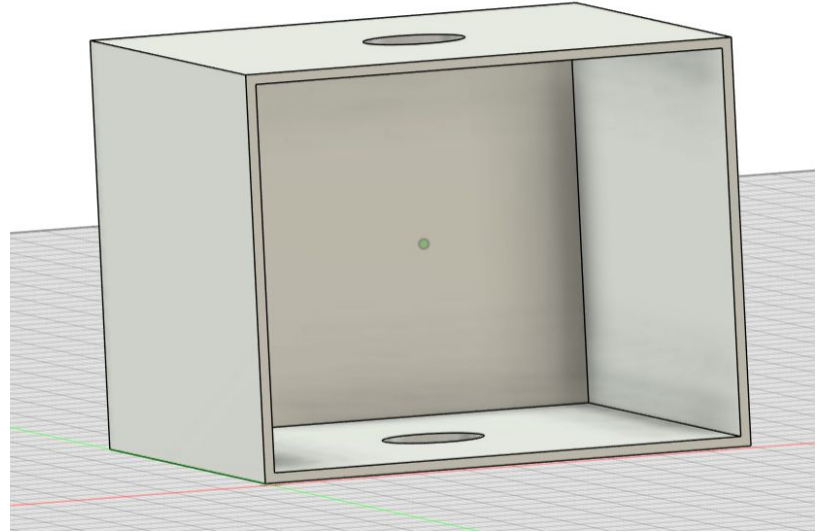
What hollows out a solid while keeping wall thickness?

a) Draft

b) Shell

c) Fillet

d) Pattern

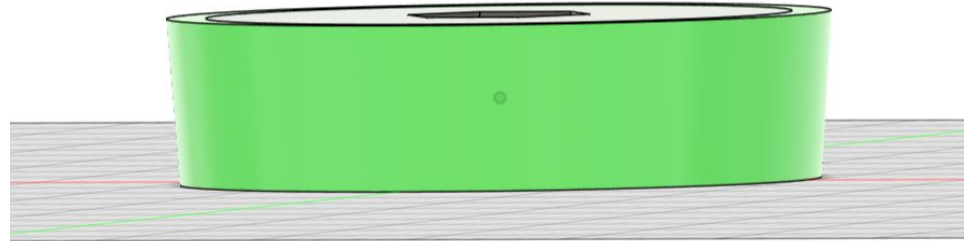
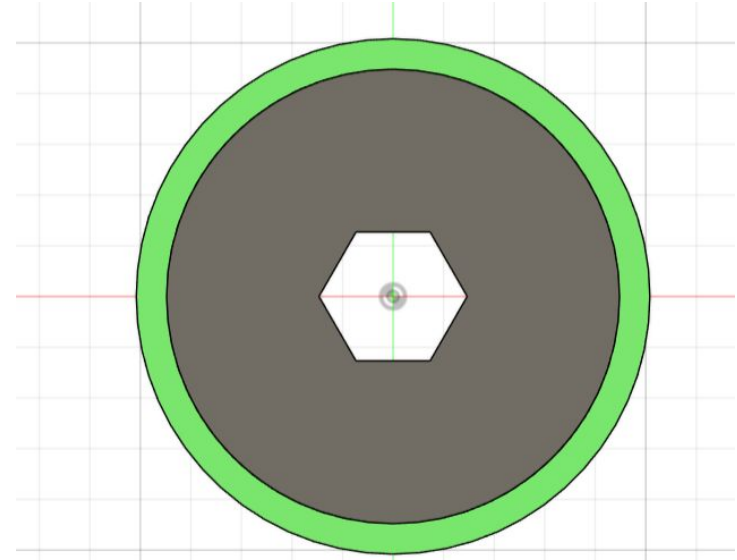


Fusion 360 Learning Check

Why apply draft to a model?

Answer: To create angled walls for easy molding or injection.

Explanation: Prevents undercuts.



Fusion 360 Learning Check

In which direction is shell thickness typically measured?

Answer: Inward (default).

Explanation: Can be outward or both for advanced designs.

